Design for Vat Photopolymerization 305

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Class Outline

Vat Photopolymerization and DFAM
Vat Photopolymerization Systems
Types of Vat Photopolymerization Processes
Vat Photopolymerization Materials
Material Types for Vat Photopolymerization
Processes and Materials Review
Vat Photopolymerization Material Considerations
Vat Photopolymerization Design Software
Design Issues for Vat Photopolymerization Parts
Vat Photopolymerization Factors for External Support Structures.
Vat Photopolymerization Factors for Internal Support Structures
Post-Processing and Finishing
Design Factors Review

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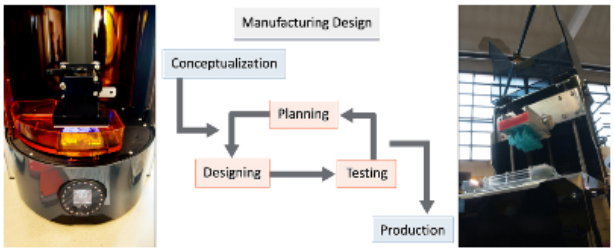
Vat Photopolymerization and DFAM



Vat photopolymerization machines use concentrated ultraviolet (UV) light to cure, or harden, liquid photoreactive resins, or photopolymers, layer-by-layer to form a solid part. Vat photopolymerization is the oldest AM technology. Because vat photopolymerization is used by desktop hobbyists utilizing consumer-level machines, it is commonly associated with the term 3D printing, though the process is also used in large industrial-level machines for end-use part production.

Although each AM method requires specific equipment, they all follow the same basic five steps of the design for additive manufacturing (DFAM) process: conceptualization, planning, design, testing, and production. The flexibility of vat photopolymerization and other AM methods enables engineers to consider things like part functionality, optimized topology, and customization. The planning, designing, and testing steps also offer AM designers opportunities to increase production efficiency. Unlike more traditional types of manufacturing, altering an AM part's design does not require additional tooling or machine configurations.

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Vat Photopolymerization Systems
Vat photopolymerization systems can vary in how they build parts. Some systems may build objects from the bottom up, which requires an optical window in the bottom of the tank to allow light and small amounts of oxygen into the vat. The small amount of oxygen creates a thin air pocket, or deadzone, right above the optical window. The oxygen deadzone prevents the liquid resin from curing in that space, which keeps the part from sticking to the vat floor. During the build process the platform moves incrementally upward, drawing the part out of the vat as it completes each layer. Other machines may build objects from the top down, which requires a light source above the vat curing each layer as a build platform incrementally lowers. Once a layer is completed, a recoater uses a blade to smooth a new coat of liquid resin over the previous layer. Differences in component and arrangement can affect the quality, cost, and speed of the vat photopolymerization process.

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Types of Vat Photopolymerization Processes
SLA, one of the original AM processes used in manufacturing, projects light from a single-point source onto the surface of the liquid photopolymer. When the light hits the liquid resin, it hardens. The light moves in a precise pattern, tracing out the part's shape and eventually creating a solid part layer.

DLP uses a specialized projector that displays a part layer as an entire image on the liquid photopolymer resin. Instead of a single-point source, DLP projectors generally have a thousand tiny light sources focused on the build platform. Upon contact with the projected light, the resin cures and forms a solid part layer all at once.

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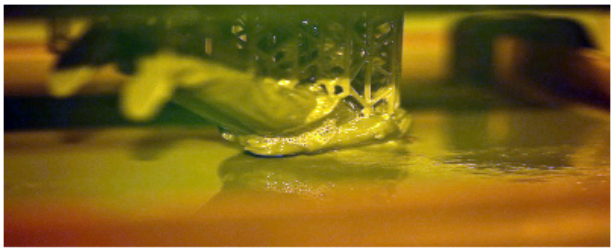
Vat Photopolymerization Materials



Vat photopolymerization often uses a specialized liquid form of thermoset plastic photopolymer resin, which permanently hardens when exposed to UV light. The fluid nature of this type of photopolymer enables the build platform to move easily inside the vat for SLA systems as the machine builds the part layer-by-layer. Though they are always liquid, materials used in vat photopolymerization systems can vary beyond plastics. For example, some systems can use photopolymer-based composites, which are infused with ceramic powders. Composites allow manufacturers to build parts with specific or enhanced properties, such as increased hardness.

Most resin types used for vat photopolymerization are proprietary, or made specifically proprietary, or designed by a specific company for use only with its own systems. Sometimes resins are designed to imitate popular plastics, such as acrylonitrile-butadiene-styrene (ABS) or polypropylene (PP). In other instances, manufacturers may develop resins to replicate the qualities of wax for more effective castings. Although manufacturers can recycle unused liquid resin, once a UV light cures a photopolymer, it is permanently altered and cannot be melted again.

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Material Types for Vat Photopolymerization

Part Prototype

When selecting a build material for vat photopolymerization, an engineer or designer must consider several factors, including the size and type of vat photopolymerization system. Desktop printers work well with standard resins that emphasize appearance, which makes them ideal for low-cost prototyping. Standard resins produce parts that have smooth, even surfaces and are generally brittle and have higher stiffness than other materials. However, standard resins are usually quicker and cheaper to print than other resin types, which makes them ideal for iterating part concepts or rapid prototyping.

Transparent Lens

Transparent resins produce parts that are translucent or clear. Manufacturers use transparent resins to build products and prototypes for camera lenses, lighting, and consumer electronics.

Polypropylene Pipe Fitting

Large industrial vat photopolymerization machines use a wider range of materials than desktop versions. Typically, industrial machines are designed for end-use production, so tough resins that have functional properties such as strength and heat resistance are optimal. Tough

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resins for vat photopolymerization have a durability that typically mimics the properties of popular plastics used in engineering. These plastic types can include ABS and PP. Common applications include automotive components, electrical housings, pipe fittings, and objects with single-snap features.

Reusable Master Patterns

Designers may use castable resins as a cheaper alternative to traditional casting and moldings. For example, an engineer may use vat photopolymerization to create master patterns for investment casting, which can create intricate shapes using a a single-use pattern or mold, which cannot be reused again. In contrast, engineers may also use heat-resistant resins to build a reusable master pattern, which can make multiple molds. Common applications can include aerospace components, molds and cores, or even jewelry.

Biocompatible Hearing Aid

Biocompatible resins, which are compatible with living tissue, allow medical and dental professionals to use vat photopolymerization to create parts with intricate features. Vat photopolymerization's ability to produce a high level of detail makes it suitable for building customizable medical implants, dental prosthetics, and hearing aids.

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Processes and Materials Review

- **Biocompatible**: A material that can be used in living tissue without being toxic or causing injury.
- **Thermoset Photopolymer**: A type of plastic resin that permanently hardens when exposed to UV light.
- **Stereolithography**: One of the original additive manufacturing techniques.
- **Vat Photopolymerization System**: Includes a movable platform, liquified resin, a light source, and a device that controls the light.
- **Design for Additive Manufacturing**: Includes the basic manufacturing steps of conceptualization, planning, and production.

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Vat Photopolymerization Material Considerations

Manufacturers must carefully consider environmental factors for storing and using materials. Materials used with desktop printers generally have fewer environmental factors compared to materials designed for industrial use.

Exposure to light can degrade photopolymer resins, which can impact repeatability, or the ability to produce consistent and uniform results. Temperature fluctuations can also affect the quality of a resin, causing it to clump in some situations. Best practices for resin storage include a steady temperature and no exposure to light.

Resins used with SLA and DLP machines are similar, but not always compatible across machines. For example, a traditional single-laser SLA printer slowly traces the build area, so a resin that cures slower may be necessary to create accurate part layers. In contrast, a DLP machine prints an entire layer at a time, which requires a faster-curing resin. When light penetrates a resin too deeply, it may cure unintended areas and affect the dimensions and details of a part. When too little light or heat causes a resin to under-cure, it may affect layer adhesion. Designers should consult the resin data sheet to ensure compatibility with vat photopolymerization systems and processes.

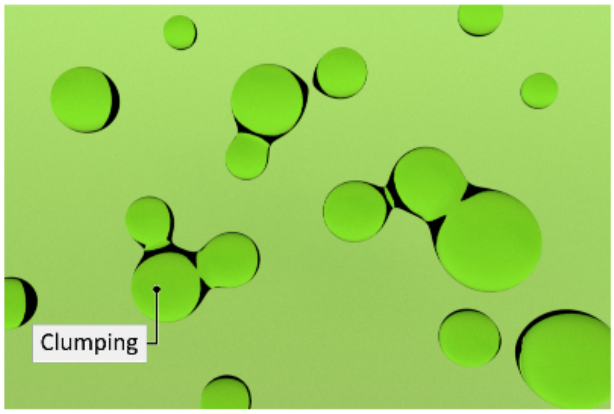

Figure 1:

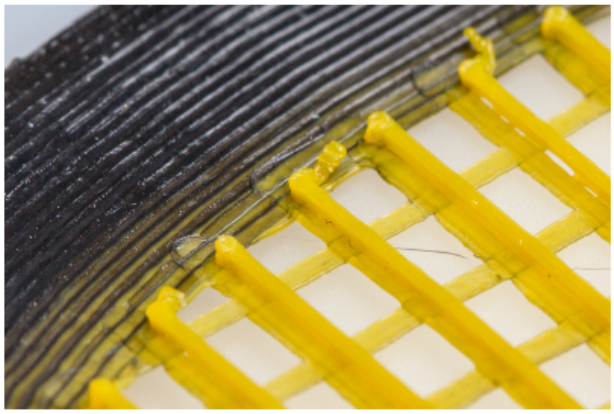

Figure 2: Layer Adhesion

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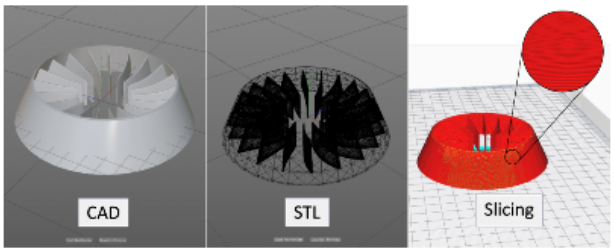
Vat Photopolymerization Design Software



However, AM machines cannot directly read design files. Because of this, a designer must convert the 3D design file into an STL file format. STL software converts the geometry of a 3D model into a sequence of interconnected triangles that represent the surfaces of the physical part. Next, digital slicer software "slices" the STL file, or translates it into digital versions of the layers that the machine will cure to physically build the part.

The digital slicer software then converts the layer information into G code, a numerical programing language that controls the functions and motions of machine tools like vat photopolymerization machines. Essentially, digital slicer software breaks a digital model into the layers of material that the vat photopolymerization machine will cure, while the G code program directs the light projection components and platform movements.

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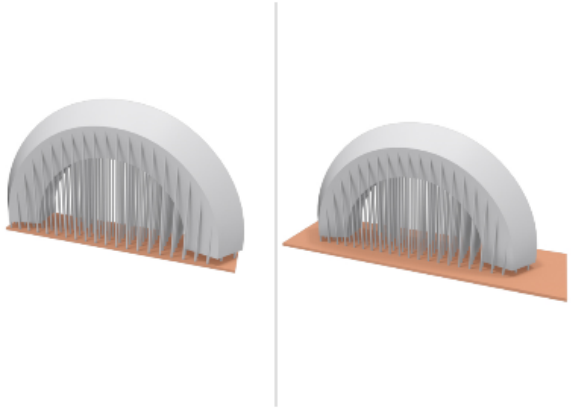
Design Issues for Vat Photopolymerization Parts

Part detaching from build platform.

Designers should take special care with the first layer of a vat polymerization part. Typically, an operator must calibrate a vat photopolymerization machine to balance the exposure time and light resolution with the part's layer width. Incorrectly calibrating may lead to under-curing, which can cause a layer to detach from the build platform or even fall off. To avoid this, some designers will expose the first layer to UV light longer to ensure adhesion.



Incorporating a wider first layer.

If the first layer of a part has a small build area, it may create design issues. When using a bottom-up system, the weight of the part may cause it to break away from the build platform during the part build. To avoid this, designers may incorporate a wider first layer to better anchor the part on the build platform.

Layer curling may cause internal part stress.

Layer warping, or curling, presents another design issue for vat photopolymerization. The energy from the UV light source causes each part layer to shrink slightly. If too much shrinkage occurs, it may cause internal stresses within the part. Designing a part with proper internal and

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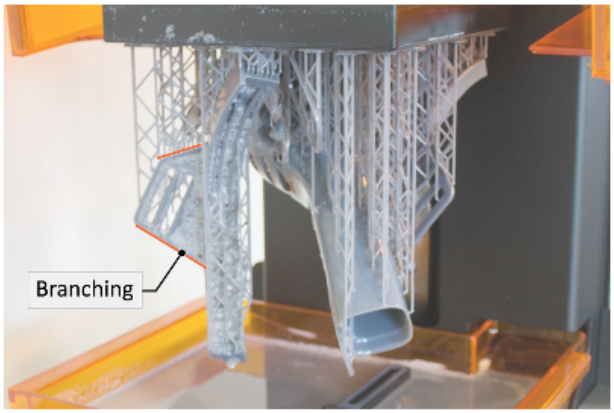
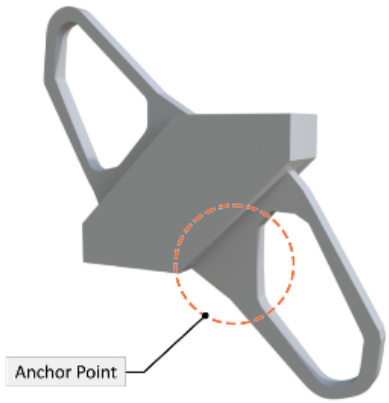
external support structures may help alleviate part stress and ensure the entire part holds its shape during the curing process.



Branches pointing outward.

If a part design has branched structures, the intersection points where two branches meet may crack or even break. Using a rigid resin type may help alleviate problems associated with intersection breakage. Optimizing the position of a part during printing, or part orientation, may also help prevent this issue. For example, by positioning the branches outward allows the intersection point to act as an anchor for the part.



Anchor points can help support lower part components.

It is also important to adequately support the base or lowest component of a part. Under-supporting the lowest point in a part structure can lead to breakage and warping during the vat photopolymerization process. To present potential defects, a designer may need to add a stronger anchor point where support structures and the main area of the part meet.

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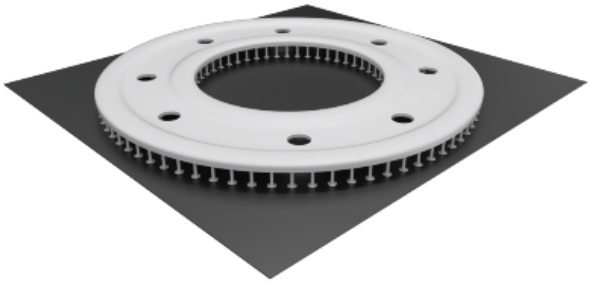
Vat Photopolymerization Factors for External Support Structures.



Flat prints need limited external support.

How a designer uses external structures depends on the vat photopolymerization system. For example, in a top-down system, a part will need external support structures for any bridges or overhangs that extend beyond a 30° angle. However, a top-down system can print flat parts with limited external support.

Angling parts reduces stress on cross-sectional areas.

In contrast, printing a flat part in a bottom-up system can lead to part stresses. A designer may therefore use support structures to angle a part, which reduces stress on its cross-sectional area during printing. This significantly reduces the risk of breakage and curling.

Printing on an incline can increase the quality.

Designers may also use support structures to put a part on an incline, which can significantly increase the quality of the printing. In addition, placing the longer support structures nearest to the recoater blade can increase the print speed, since the blade has less distance to travel on the machine.

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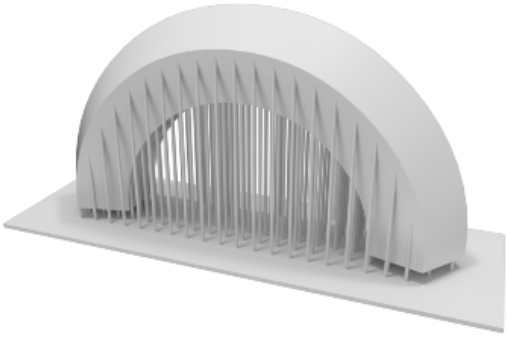
Interconnected support structures provide sturdier supports.

In some cases, designers may interconnect support structures to provide sturdier support for heavier parts. However, in general the additional resin used to create support structures increases the overall cost of the part.



Avoid using support structures on visible parts.

To avoid extensive post-processing, vat photopolymerization designers orientate the part so that external support structures are used in the least visible part areas, which prevents support blemishes in the most visible areas of a part.

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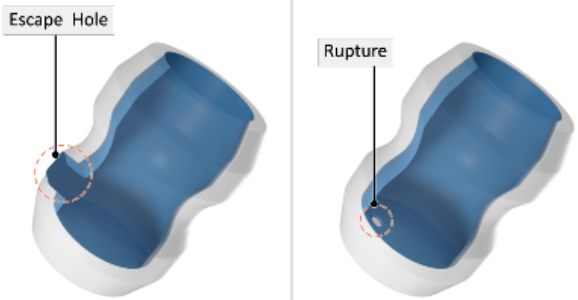
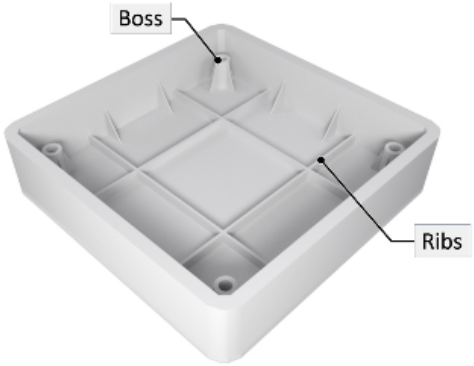
Vat Photopolymerization Factors for Internal Support Structures



Adding escape holes can reduce cupping.

In order to reduce material costs and decrease print time, vat photopolymerization designers may make the internal structures of non-functioning prototypes and parts hollow. However, liquid resin trapped in these hollow areas may cause the part to become unstable during the printing process. Bottom-up systems in particular may experience a type of part rupture referred to as cupping. In cupping, liquid photopolymer becomes trapped within a hollow part, which creates a suction-like force that can rupture part walls during the build process. Adding escape holes to the part design, which allow the liquid to flow out of the part, can reduce the risk of cupping.



Internal support structures can reduce stress fractures.

Designers may use a variety of internal support structures to reinforce the thin walls of a hollow part or help join pieces together. For example, a raised, circular protrusion with a hole in the center, called a boss, may guide a screw or pin of a mating part to help improve part assembly. Designers may also add a support rib to increase the stiffness of a part and help reduce potential warping. Using rounded part corners instead of sharp corners may also help reduce the potential for stress fractures.

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Post-Processing and Finishing

After printing, most vat photopolymerization parts will need post-processing or finishing procedures to clean, improve, or create an attractive surface finish on a part. External support material must be removed, which requires specific techniques to avoid damaging any delicate part features. Once parts are printed, an operator typically removes support structures by hand. Often, completed vat photopolymerization parts are cured again to further solidify and strengthen the part.

Post-processing can improve surface finish and remove blemishes or imperfections that may have occurred during the vat photopolymerization process. Because post-processing may add material to or remove material from a part, designers must factor it into their part designs. For example, sanding and other abrasive finishing processes remove small amounts of material from the part's surface. If a part needs to be filed or sanded, a vat polymerization designer should build the part a little larger to account for the removed material. Painting or adding other coatings increases a part's size slightly, so in these instances, the designer should design the part slightly smaller to account for the additional material. Other parts may present multiple finishing or post-processing options to achieve different affects. For example, operators might apply a clear coating to a transparent surface to give it a more vibrant finish. In some instances, adding a mineral oil coating can also help give a part an even finish.

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Design Factors Review

Internal Support Structures

- Rib
- Hole

Design Issues

- Curling
- Intersection Break
- Undercuring

Post-Processing

- Painting
- Mineral Oil Coating

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Class Vocabulary

3D Printing	The process of producing a 3D object using a specialized tool that creates successive layers of material. 3D printing must technically involve the use of a print head or nozzle, but the term is often used interchangeably with additive manufacturing (AM).
ABS	Acrylonitrile-butadiene-styrene. A thermoplastic material with good heat-, chemical-, and impact-resistance. ABS is commonly used to manufacture medical equipment housings.
Acrylonitrile-Butadiene-Styrene	ABS. A thermoplastic material with good heat-, chemical-, and impact-resistance. Acrylonitrile-butadiene-styrene is commonly used to manufacture medical equipment housings.
Additive Manufacturing	AM. The process of successively layering materials to make an object based on a three-dimensional computer model. Additive manufacturing allows for rapid prototyping, mass customization, and increased part complexity.
Anchor Point	Sources of security or strength on an object. Anchor points supply a secure hold or support for something else.
Biocompatible Resins	A material that can be used in living tissue without being toxic or causing injury. Biocompatible resins are frequently used with vat photopolymerization to make medical implants.
Boss	A raised, circular peg or protrusion. A boss often has a hole in the center and is used to improve assembly.
Bridges	A vertical column that holds up a horizontal part feature. Bridges are used in additive manufacturing to prevent structural sagging.
Brittle	Resistant to being drawn, stretched, or formed. Brittle materials can fracture when subjected to mechanical forces.
Calibrate	Comparing and adjusting a device with unknown accuracy to a device with a known, accurate standard. To calibrate a device requires eliminating variation for increased accuracy.
Castable Resins	A liquid photopolymer material that hardens to create an expendable patterns or molds when exposed to ultraviolet (UV) light or heat. Castable resins typically have high green strength and can be used for jewelry and investment casting.
Castings	A part formed by pouring or injecting heated material into a mold. Castings take shape as the material cools and solidifies.
Ceramics	Consists of both metallic and nonmetallic atoms held together by a strong primary bond. Ceramics, which include metal oxides, nitrides, and glasses, are sometimes used as build material in additive manufacturing.
Computer-Aided Design	CAD. Computer software used to create a 3D model of a part before it goes into prototyping or production. Computer-aided design models are digital representations of the different aspects of a part.
Conceptualization	To form an idea about a possible part or manufactured good. Conceptualization is one of the initial stages of design for manufacturing (DFM) or design for additive manufacturing (DFAM).
Cupping	Excess build material or air becoming trapped within a a hollow part causing it to warp or deform. Cupping ruptures typically occur during bottom-up stereolithography processes.
Cure	To cause a material to bond and solidify by permanently cross-linking its molecules. Curing techniques for thermosets include exposing the material to ultraviolet light during the production process.
Curling	Physically bending, twisting, or deforming. Curling can be caused by uneven heating and cooling and by a lack of reinforcement in a part component.
Design	In the five-step design for additive manufacturing process, the creation of the actual part specifications. Design involves creating the blueprints and prototypes for a part.

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Design For Additive Manufacturing	DFAM. The methodology of planning, testing, and creating an additively manufactured part that functions optimally. Design for additive manufacturing allows engineers to primarily focus on part functionality.
Desktop	A machine designed to fit easily on a standard desk. Desktop models of AM machines are more affordable but less precise than larger, more advanced models.
Digital Light Processing	DLP. A vat photopolymerization process that uses a specialized device to create an entire part layer in one flash of light. Digital light processing builds parts more quickly than some other vat photopolymerization processes.
DLP	Digital Light Processing. A vat photopolymerization process that uses a specialized device to create an entire part layer in one flash of light. DLP builds parts more quickly than some other vat photopolymerization processes.
End-Use	Designed to be used directly by a consumer or directly in another manufactured product. End-use products created by additive manufacturing (AM) include medical implants, custom dental devices, and camera equipment.
Environmental Factors	The characteristics that make up the surroundings in a given place. The workplace environment consists of various environmental components, including noise and air quality.
Exposure Time	The length of time during which light remains in contact with a thermoset. Exposure time affects the quality of curing during vat photopolymerization.
Finishing	A process applied to a manufactured part that goes beyond making the part to the correct shape and size. Sanding, filing, heat treating, and painting are examples of finishing.
G Code	A programming language that uses letters and numbers to form commands for machine tools. G code programs are used to direct tool movements in additive manufacturing, CNC machining, and hybrid machining.
Heat Resistance	The ability to resist degradation when exposed to extreme heat. Heat resistance is important for plastics that will be exposed to heat on a regular basis.
Investment Casting	A casting process that creates intricate shapes using an expendable pattern and mold. Investment casting patterns can be made using additive manufacturing processes.
Iterating	Repeating actions and steps in processes. Iterating is an important aspect of part design because it allows for constant and quick re-designs during initial stages.
Lasers	A device that generates an intense beam of light that can be precisely aimed and controlled. Lasers are used in many additive manufacturing processes.
Layer Adhesion	The measure of bond strength in additive manufacturing (AM) processes. Layer adhesion in vat photopolymerization can be affected by under-curing or over-curing.
Light-Emitting Diodes	LEDs. A semiconductor device that emits a narrow spectrum of light in a forward direction. Light-emitting diodes are sometimes used as light sources in vat photopolymerization.
Master Patterns	A replica of a part or casting that is used to create a negative impression in a mold. Master patterns for molding and casting processes are traditionally made of plastic, wood, or metal.
Mating Part	The object that a workpiece has been manufactured to fit. Nuts and bolts are mating parts.
Mineral Oil	A naturally occurring oil derived from petroleum. Mineral oil is used as a lubricant, finish, and coolant.
Overhangs	Structure on a part that extends without support beneath them. Overhangs that extend beyond a 30-degree angle typically require support structures in vat photopolymerization processes.
Part Orientation	The position a part will be manufactured in during an additive manufacturing process. Part orientation, when set properly, minimizes build time and maximizes part quality.
Photopolymers	A material that chemically reacts to exposure to ultraviolet (UV) light. Photoreactive resins solidify and harden when exposed to UV light.
Photoreactive Resins	A material that chemically reacts to exposure to ultraviolet (UV) light. Photoreactive resins solidify and harden when exposed to UV light.
Planning	A step in the part design additive manufacturing process that assesses the best ways to create the conceptualized part. Planning involves initial considerations of the design and of the practical concerns related to the manufacturing process.

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concerns related to the manufacturing process.

Polypropylene PP. A polymer that is tough, flexible, and resistant to fatigue. Polypropylene is commonly used to make translucent products such as food packaging.

Post-Processing A procedure used to clean, improve, or otherwise finish a part for use by a manufacturer or consumer. Post-processing additively manufactured parts includes abrasive finishing, heat treatment, and painting.

PP Polypropylene. A polymer that is tough, flexible, and resistant to fatigue. PP is commonly used to make translucent products such as food packaging.

Production The manufacturing of a finished part for delivery to a customer. Production is the final stage of design for additive manufacturing (DFAM).

Proprietary Designed by a specific company for use only with its own systems. Proprietary software, files, and materials are sometimes used with specific additive manufacturing (AM) machines.

Prosthetics An artificial implant, device, or limb that replaces a missing body part. Prosthetics may also refer to the design, fabrication, and fitting process involved with these devices.

Rapid Prototyping A product development technique in which additive manufacturing (AM) methods are used to create prototypes for a traditional manufacturing operation. Rapid prototyping allows engineers to quickly create a number of prototypes in a short time period, reducing lead time.

Recoater A device that slides forwards and backwards over the surface of photopolymer material held in a vat. Recoaters are used by stereolithography (SLA) systems to smooth photopolymer material layers over a build platform.

Repeatability The ability of a manufacturing process to produce uniform results. Repeatability enables the production of parts with a high degree of consistency.

Resolution The amount of pixels and detail in an image. High-resolution images have better details than lower-resolution images.

Rib A structural component that provides shape and support. Ribs are used to reinforce a thin section of a part, such as a wall.

Single-Snap Features Interlocking component, often made out of metal or plastic, that easily fasten together when force is applied. Single-snap features can simplify assembly since they are fastened by one quick application of pressure.

SLA Stereolithography. A vat photopolymerization process that traces a focused light beam over a layer of photopolymer to selectively cure it. SLA is often used for making softer parts that require a high degree of customization, such as dental aligners or hearing aids.

Slicer A computer program that divides an STL file into layers for additive manufacturing (AM). Slicers, or slicing programs, can either be separate computer programs or part of the machine interface.

Software Coded instructions or programs that control computer hardware functions and operations. Software is used to design additively manufactured parts and command machine tool movements during part creation.

Standard Resins A liquid photopolymer material that becomes stiff and brittle when exposed to ultraviolet (UV) light or heat. Standard resins have a smooth surface and are typically low-cost, which makes them ideal for prototyping with vat photopolymerization.

Stereolithography SLA. A vat photopolymerization process that traces a focused light beam over a layer of photopolymer to selectively cure it. Stereolithography is often used for making softer parts that require a high degree of customization, such as dental aligners or hearing aids.

Stiffness The ability of a material to resist bending or stretching. Stiffness is sometimes called rigidity.

STL Software that can create or convert digital design files, which represent 3D models, into a series of interconnected triangles. STL software is used in additive manufacturing processes.

Strength A material's ability to resist forces that attempt to break or deform it. Additive manufacturing processes such as vat photopolymerization create parts with poor strength.

Stress Fractures A hair-like crack that appears on the surface of a part when the material has been exposed to excess tensile or shear stress. Stress fractures often lead to part failure.

Support Structures A reinforcing component used to hold the weight of an additively manufactured part as it is being

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Support Structures A reinforcing component used to hold the weight of an additively manufactured part as it is being constructed. Support structures are removed from the part once the build is complete.

Testing In the five-step design for additive manufacturing process, examining a part to ensure that it performs its intended function and that it can be satisfactorily manufactured. Testing indicates whether the part needs additional planning or if it is ready for production.

Thermoset A group of plastics that are permanently hardened by light or heating. Thermosets can be used as build materials in some additive manufacturing methods, such as vat photopolymerization and material jetting.

Topology The arrangement of geometric or material components in a part. The topology of additively manufactured parts is particularly complex because it is capable of creating intricate shapes and various gradients of material in a part.

Tough Resins A liquid photopolymer material that becomes tough and impact-resistant when exposed to ultraviolet (UV) light or heat. Tough resins are often used with vat photopolymerization to imitate the capabilities of engineering-grade plastics.

Translucent Able to permit the passage of light without being fully clear. Translucent materials are used in additive manufacturing to make parts used in fields such as optical, lighting, and consumer electronics.

Transparent Resins A liquid photopolymer material that becomes clear or translucent when exposed to ultraviolet (UV) light. Transparent resins used with vat photopolymerization can have varying degrees of transparency.

Ultraviolet UV. A potentially harmful wavelength of light that is below, or shorter than, violet on the light spectrum. Ultraviolet light is used to selectively harden a photopolymer in vat photopolymerization processes.

UV Ultraviolet. A potentially harmful wavelength of light that is below, or shorter than, violet on the light spectrum. UV light is used to selectively harden a photopolymer in vat photopolymerization processes.

Vat Photopolymerization An additive manufacturing (AM) method that builds a part by curing layers of photopolymer with light. Vat photopolymerization processes include stereolithography (SLA) and digital light processing (DLP).

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